

OLOWU ABRAHAM ADURAGBEMI

Data Analyst | SQL · Python · Power BI · Business Intelligence · LLM Agent Engineering

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PROFESSIONAL SUMMARY

Data Analyst and validation specialist with experience across fintech, healthcare, pharmaceutical, and retail domains. Builds dashboards, monitors KPIs, and translates complex datasets into clear business answers that help decision-makers act with confidence. Produces written analytical summaries that strip ambiguity from findings before they reach the decision desk. At Stratum Core Systems, independently designs and executes structured validation frameworks, auditing analytical outputs against release criteria and documenting findings with the precision required for executive-level review. Work spans fraud pattern analysis, statistical modeling, LLM agent engineering, and performance reporting across datasets exceeding 600,000 transactions and 110,000 records. Proficient in SQL, Python, Power BI, Tableau, Excel, and the Claude API.

TECHNICAL SKILLS

Analysis & Reporting: KPI Monitoring, Trend Analysis, Cohort Analysis, Anomaly Detection, Hypothesis Testing, Statistical Modeling, Time Series Forecasting, EDA

BI & Visualization: Power BI, Tableau, Excel (Advanced Formulas, PivotTables, Dashboards), Streamlit, Matplotlib, Seaborn

Programming & Databases: Python (Pandas, NumPy, Scikit-learn, XGBoost, SHAP), SQL (PostgreSQL, MySQL, CTEs, Window Functions, Query Optimization), TypeScript

Validation & Governance: Data Validation, Mutation Testing, Hash-Based Integrity Verification, Structured Reporting, Documentation, ETL, Data Cleansing

AI & Agent Engineering: Claude API (Anthropic), LLM Prompt Engineering, Multi-Model Orchestration, RAG Architecture, Prompt Chaining, Zod Output Validation, Agent Pipeline Design

Tools & Platforms: Google BigQuery, Jupyter Notebooks, Git, GitHub, Next.js 15, Supabase, Vercel, Google Analytics, Google Sheets, Excel Macros, Stripe, Paystack

PROFESSIONAL EXPERIENCE

Founding Analytical Associate / Data Analyst

Stratum Core Systems Inc. — Lagos, Nigeria (Remote) | March 2026 — Present

- Rebuilt a data-completeness calculation from its written specification and reconciled it against a frozen reference dataset, verifying every record across two datasets (10,138/10,138 and 10,144/10,144 matched) and confirming the validation checks themselves were sound using mutation testing — deliberately injecting a fault to prove a real defect would be caught
- Identified and quantified a calculation-logic defect in an already-approved release, where a validation threshold had been applied to a full time period instead of each half-period as specified — measuring the impact as 24 of 144 records (roughly 1 in 6) that passed under the shipped logic but failed the documented rule
- Independently verified a corrected business-rules module against all 6 acceptance criteria, caught a defect the previous submission had missed (24 output-classification changes reported as zero), and reproduced the correct figure three independent ways, including a bit-for-bit rerun of the code from source
- Validated a six-revision analytical calculation engine against its approved specification, reproducing 45 file-integrity hashes, 22 acceptance test vectors, and 30 unit tests, then built 9 additional cross-check tests through a separate code path, surfacing a confirmed 1,356-record discrepancy between two implementations plus two requirements ambiguities escalated to the product owner
- Caught a denominator miscalibration that inflated exposure estimates by over 15,000% against a 40% instability threshold, blocking an unusable analytical run from executive review until resolved
- Built independent Python validation frameworks and controlled governance registers (with SHA-256 hash-based version tracking) covering scenario design, defect tracking, and evidence indexing across multiple release cycles
- Produced structured executive-facing readiness notes per analytical run, writing summaries that separate what the data shows from what it means, so leadership can act on findings without a technical interpreter in the room

Data Analytics Consultant (National Service)

Government Technical College — Osogbo, Nigeria | February 2025 — December 2025

- Built 3 regression models quantifying the relationship between student attendance and academic outcomes across 120+ records
- Designed a performance monitoring framework tracking 15 academic indicators for early identification of at-risk students; class average improved 15%

Digital Community Growth Analyst

Freelance — Remote (United States) | January 2022 — March 2023

- Tracked 71,500+ members across 2 online communities, analyzing participation patterns for growth opportunities
- Planned and evaluated 4 A/B experiments on content timing and format using statistical significance testing
- Monitored DAU/MAU ratios and retention metrics weekly via Google Analytics, compiling findings into Excel dashboards
- Recommended scheduling and format changes that produced a 25% increase in community participation

Quality Assurance Analyst

Archy Pharmaceutical Company — Lagos, Nigeria | June 2021 — October 2021

- Monitored 200+ pharmaceutical batch tests via statistical process control charts, maintaining 100% GMP compliance
- Flagged 15+ out-of-spec patterns before escalation, contributing to a 20% reduction in batch rework incidents

Inventory Analyst

Preplus Supermarket — Lagos, Nigeria | March 2020 — January 2021

- Built demand forecasting models across 500+ SKUs using time series analysis; stockouts dropped 30%, inventory turnover improved 15%
- Automated reorder triggers from forecasted demand thresholds, cutting manual purchasing review time by 40%

PORTFOLIO PROJECTS

BetGenius — AI Prediction and Agent Behavior Platform

2025 | Next.js 15 · TypeScript · Supabase · PostgreSQL · Paystack

- Shipped a 3-tier SaaS product solo covering full architecture from schema design to production deployment
- Built a 2-pass LLM agent pipeline with autonomous model selection and rate-limit-protected sequential predictions

- Crafted 6 behavioral rules into the AI's persona via system prompts for confidence-tiered, uncertainty-flagged output
- Validated all AI outputs through Zod schemas before storage, ensuring zero malformed data reaches users

The Last 40 Million — Statistical Drivers of Financial Exclusion in Nigeria

2024 | Python · statsmodels · SciPy · Pandas

- Analyzed 26,930 Nigerian adults across 6 inferential statistical modules using nationally representative survey data
- Quantified a 34x exclusion odds ratio for adults with no formal education vs university graduates (OR = 34.15, 95% CI [27.85, 41.88])
- Corrected a female-skewed income selection bias affecting 4,924 respondents; published a 14-page findings report and live dashboard

Fintech Transaction Intelligence — Merchant Risk Analysis

2024 | Python · SQL · XGBoost · SHAP · Streamlit (synthetic dataset)

- Analyzed 600,000+ transactions across 5,000 merchants (NGN 26.3B volume) to flag 973 high-risk merchants via XGBoost with SHAP
- Engineered 40+ risk features and 23 SQL queries; deployed a 4-page interactive Streamlit dashboard

Patient Retention — No-Show Prediction in Healthcare

2024 | Python · SQL · Random Forest · XGBoost · SMOTE (Kaggle no-show dataset)

- Built a stacking ensemble at 0.84 test ROC-AUC on 110,527 medical appointment records (62,299 patients); segmented patients into risk tiers
- Modeled \$2.37M in projected revenue recovery (2,149% first-year return) from preventing 13,000+ annual no-shows

Nigerian Fintech Marketing Channel Analysis

2024 | Python · Plotly · MySQL · Power BI (synthetic campaign data)

- Analyzed 500 synthetic ad campaigns across 10 industries; Google Ads delivered a \$0.74 CPA vs \$2.09 on Facebook, with Facebook leading on audience reach and ARPU
- Delivered a 3-page interactive Power BI dashboard tracking ROI, CPA, and ARPU across platforms

Customer Churn Prediction — Telecom Analytics

2024 | Python · Scikit-learn · Random Forest · SMOTE · SHAP

- Built an 84.2% ROC-AUC churn model for 7,043 subscribers; quantified \$1.73M in annual revenue at risk
- Designed a retention program projecting a 411% first-year ROI and \$577K revenue protected

Retail Sales Intelligence — Revenue Optimization

2024 | MySQL · Python · Matplotlib · Seaborn

- Unlocked \$262K Year 1 net benefit (577% ROI) via BI analysis of 2,000 transactions (\$766K revenue)
- Traced 40.5% of revenue to Electronics and identified the peak selling hours and days driving daily sales

Nexus Inventory — Multi-Tenant SaaS POS Platform

2025 | Next.js 15 · Supabase · Drizzle ORM · Stripe · Vercel

- Built and launched 5 core modules solo, covering complete architecture, multi-tenant auth, and deployment
- Listed for acquisition; demonstrates end-to-end product ownership as sole developer

EDUCATION

Bachelor of Science — Biochemistry

Federal University Oye Ekiti (FUOYE) — Ekiti, Nigeria | September 2018 – August 2024

CERTIFICATIONS

- Advanced Learning Algorithms | DeepLearning.AI / Stanford University (Coursera) | 2026
- Supervised Machine Learning: Regression & Classification | DeepLearning.AI / Stanford University (Coursera) | 2026
- Understanding and Visualizing Data with Python | University of Michigan (Coursera) | 2026
- Data Analytics Job Simulation (Tableau, Excel, Forensic Analytics) | Deloitte via Forage | 2026
- Foundations of Business Intelligence | Google (Coursera) | 2026
- SQL and Relational Databases 101; Python 101 for Data Science | IBM (Cognitive Class) | 2026
- Introduction to Data Science | Cisco Networking Academy | 2026